Midterm review session

GOVT 6039 · Statistical and Computational Models · Fall 2026

Coefficient interpretation, maximum likelihood and Bayesian inference

Instructor: Daniel de Kadt · Session led by Georgy · **Instructor version (answers in red)**

The midterm. Tuesday, October 27, in class, 75 minutes. It covers Units 1 and 2 (Lectures 1–14). You may bring two pages of notes and no devices.

Contents. Part I covers how to interpret a coefficient in each model from the course. Part II compares maximum likelihood with Bayesian inference. Each part starts with a summary of the main idea and the tools, followed by exercises in the style of the exam. Appendix A is a one-page summary that you can use as one of your two pages of notes.

Using the handout. Each exercise begins with a setup, which gives the model, the data and the numbers you need, and a task. Try the exercises before reading the answers. Most of the calculations can be done by hand with the pocket table (Table 3) and the values in Appendix A.

Notation and abbreviations. Notation follows the lectures: $\eta_i = x_i^\top \beta$, $g\{\mathrm{E}[Y_i \mid x_i]\} = \eta_i$, $L(\theta; Y) = p(Y \mid \theta)$, $\hat{\pi}(x) = \mathrm{logit}^{-1}(x^\top \hat{\beta})$. Lecture numbers appear in parentheses, so (L8) refers to Lecture 8. pp is percentage points, lgdp is the natural log of GDP per capita, OR is odds ratio, IRR is incidence rate ratio, PEA is the partial effect at the average, APE is the average partial effect, and ML and MLE are maximum likelihood and the maximum likelihood estimate.

Data used in the examples. Most numbers in this handout come from models estimated in lecture. The regime models (L7–L10, L12) use cross-national data with one row per country. The two covariates are oil, coded 1 for oil-producing states and 0 otherwise, and lgdp, the natural log of GDP per capita. One unit of lgdp is a 2.72-fold difference in income, and doubling income adds $\log 2 = 0.69$ to lgdp. The outcome differs by model: democracy coded 0/1 (LPM, logit and probit; L7–L8), regime type ordered from closed autocracy to electoral autocracy to democracy (ordered logit; L9), type of autocracy, which is military, single-party or personalist (multinomial logit; L9), the number of protests per year (Poisson; L10), and the number of years before a regime ends (discrete-time hazard; L12).

The Bayesian examples (L13–L14) use a survey of trust in local government in Chinese provinces, where $Y_i = 1$ if respondent i trusts local government. Hainan has $n = 15$ respondents, 12 of whom say yes, and Guangdong has $n = 236$. The parameter π is the share of all adults in the province who trust local government. Three examples are not from lecture and were chosen to keep the arithmetic simple: the turnout logit in Exercise 3, the baseline survival of 0.50 in Exercise 7 and the four-respondent province in Exercise 11.

Overview

Both review questions concern what an estimate from a fitted model means and how uncertain it is. Part I deals with the meaning of $\hat{\beta}$ in each model. Part II deals with two ways of describing uncertainty about an estimate. The box below states the main point of each part.

Part I. In every model in the course, a one-unit increase in x_{im} with the other covariates held fixed adds β_m to the linear predictor $\eta_i = x_i^\top \beta$. The models differ in how a change in η_i translates into a change in the outcome, and the link function g determines this. With the identity link, β_m is the change in the mean of Y. With any other link, the change in the outcome depends on the starting value of η_i, so it has to be computed as a predicted quantity.

Part II. Maximum likelihood and Bayesian inference use the same likelihood, $L(\theta; Y) = p(Y \mid \theta)$. ML reports the value of θ that maximizes L and measures uncertainty by how much $\hat{\theta}$ would vary across repeated samples. Bayesian inference multiplies L by a prior $p(\theta)$ and reports the posterior distribution $p(\theta \mid Y)$, which describes uncertainty about θ given the observed data. The prior is an assumption that has to be justified, and its influence declines as n grows. Posterior draws can be used to compute predicted quantities in the same way as the simulation method in Part I.

The plan for the 75-minute session is below. Material not covered in the session is written out in full in the handout.

Minutes	Sections	Content
0–5	Overview	main points of Parts I and II; interpretation steps (§1)
5–15	§1–2	Table 1 and Figure 1: odds ratios and changes in probability
15–45	§3	Exercises 1–8: interpretation and calculation
45–60	§4–6	ML and Bayesian inference for Hainan: Tables 4–5, Figures 3–4
60–72	§7–9	Exercises 9–15: priors, grid, Metropolis step, intervals
72–75	Wrap-up	summary sheet (Appendix A)

Part I Interpreting coefficients across models

1 Coefficients and the linear predictor

Every model in Units 1–2 has three parts (L7): a distribution for Y_i (normal, Bernoulli, Poisson and so on), a linear predictor η_i, and a link function g that connects the mean of Y_i to η_i,

$$\eta_i = x_i^\top \beta, \qquad g\big(\mathrm{E}[Y_i \mid x_i]\big) = \eta_i \quad \Longleftrightarrow \quad \mathrm{E}[Y_i \mid x_i] = g^{-1}(x_i^\top \beta).$$

A one-unit increase in x_{im}, with the other covariates fixed, changes η_i by β_m. Two consequences follow.

1. On the scale of η, the effect is the same for every unit and every value of x. This is why the log-odds lines in Figure 1 (left) are parallel and why the odds, rate and hazard ratios in Table 1 are constant. In the logit of democracy, oil lowers the log odds by 1.56 for poor and rich countries alike.
2. On the scale of $\mathrm{E}[Y_i \mid x_i] = g^{-1}(\eta_i)$, the effect equals β_m only when g is the identity. For other links g^{-1} is curved, so the same change in η produces a larger or smaller change in the outcome depending on the starting value of η_i, and therefore on all of x_i. The change of -1.56 in log odds lowers the probability of democracy by 8 percentage points for a country starting at $\pi = 0.10$ and by 33 points for a country starting at $\pi = 0.50$ (Table 2).

The link keeps $\mathrm{E}[Y_i \mid x_i]$ within the range allowed by the outcome. Probabilities must lie between 0 and 1 (logit, probit and cloglog links), and expected counts must be positive (log link). A linear function cannot satisfy these limits, so the model is linear in η and nonlinear on the scale of the outcome. Because the slope of a nonlinear function changes along the curve, a single coefficient implies different effects on the outcome for different units.

> **Steps for interpreting one coefficient** $\hat\beta_m$ (L8, L9, L10, L12). We interpret the estimate $\hat\beta$, not the estimand β, and hold the other covariates fixed.
>
> 1. Read the sign. Outside the simplest models the sign does not always carry over to the outcome. The exceptions are the middle categories of an ordered outcome and any category of a multinomial outcome.
> 2. Exponentiate. $e^{\hat\beta_m}$ multiplies the odds (logit and ordered logit), the expected count (Poisson and negative binomial) or the hazard (cloglog). A multiplier of 0.21 is a 79% reduction, and a multiplier of 1.88 is an 88% increase.
> 3. For a change of size c, the multiplier is $e^{c\hat\beta_m}$. Doubling GDP per capita corresponds to $c = \log 2$, so the multiplier is $e^{\hat\beta_m \log 2} = 2^{\hat\beta_m}$.
> 4. If the reader needs probabilities, counts or survival, compute a predicted quantity (§2).
>
> Steps 2 and 3 follow from the link. Exponentiating reverses the log, so an additive change in η becomes a multiplicative change in the odds, the expected count or the hazard. As L8 notes, other GLMs require different versions of steps 2 and 3, and even step 1 may fail.

Table 1 applies these steps to each model in the course. The second column gives what $\hat\beta_m$ changes, the third gives what $e^{\hat\beta_m}$ multiplies, and the fourth says whether a predicted quantity is needed to describe the effect on the outcome.

Table 1: Interpretation of a one-unit increase in x_{im} in each model. Lecture numbers in parentheses.

Model, link g	$\hat{\beta}_m$ is the change in…	$e^{\hat{\beta}_m}$ is…	Predicted quantity needed?
OLS (L2) identity	$\hat{\mu}$, in units of Y, the same at every x. FWL: the slope of residualized Y on residualized x_m.	—	No. $\hat{\beta}_m$ is already the change in $\mathrm{E}[Y \mid x]$. (It is the slope of the best linear predictor $\beta^\dagger$, not necessarily of the true curve.)
Linear probability (L7) identity	$P(Y_i = 1 \mid x_i)$, in probability points; $\times 100$ gives percentage points.	—	Rarely. A constant slope cannot hold near 0 and 1, and 27 of 205 fitted "probabilities" fall outside $[0, 1]$.
Logit (L7–8) logit	the log odds of $Y_i = 1$.	the odds ratio: multiplies the odds $\pi_i/(1-\pi_i)$.	Yes, for any statement about probability: $\hat{\pi}(x)$, a contrast $\hat{\tau}_j(x)$, a PEA or APE. Slope: $\hat{\beta}_j \hat{\pi}(1-\hat{\pi})$.
Probit (L7) Φ^{-1}	the index $x_i^\top \beta$ (in standard-deviation units of the latent normal).	no odds-ratio reading	Yes, always. $\hat{\beta}$ is on a different scale from the logit coefficients, so compare $\hat{\pi}$ rather than $\hat{\beta}$.
Ordered logit (L9) cumulative logit	the log odds of being above any cutpoint ζ_j (proportional odds). The latent distribution shifts right by $\hat{\beta}_m$, and the $\hat{\zeta}_j$ do not change.	multiplies the odds of being above any cutpoint.	Yes. $P(Y_i \le j \mid x_i) = \mathrm{logit}^{-1}(\zeta_j - x_i^\top \beta)$. Only the first and last categories must follow the sign, and the three APEs sum to 0.
Multinomial logit (L9) baseline-category, $\beta_1 = 0$	the log odds of category j *relative to the base* category.	multiplies $P(Y_i = j \mid x_i)/P(Y_i = 1 \mid x_i)$.	Yes. $\hat{\beta}_{jm}$ does not determine the change in $P(Y_i = j \mid x_i)$, and a negative coefficient can occur with a positive APE.
Poisson, NB (L10) log	the log expected count.	the incidence rate ratio: multiplies $\mathrm{E}[Y_i \mid x_i]$, for every unit.	For counts. The difference is $\hat{\mu}(x)(e^{\hat{\beta}_m} - 1)$, which grows with $\hat{\mu}$, while the ratio is constant.
Discrete-time hazard (L12) cloglog	$\log\{-\log(1-h_{ij})\}$, the same in every spell year j (proportional hazards).	the hazard ratio: multiplies the hazard rate λ_{ij} ($\approx h_{ij}$ when small).	Yes, for survival: $\hat{S}(t \mid x) = \prod_{j \le t}\{1 - \hat{h}_j(x)\}$. The hazard ratio is not a ratio of survival probabilities.
Hierarchical logit (L14) logit, intercept $\beta_{0j} = \beta_0 + \beta_{\mathrm{lgdp}}\mathrm{lgdp}_j + u_j$	the log odds with u_j held fixed, i.e. for respondents in the same province.	odds ratio, within a province.	Yes, for provincial levels: $\hat{\pi}_c = \mathrm{logit}^{-1}(x_c^\top \hat{\beta} + \hat{u}_j)$, then poststratify (MRP).
Ridge, lasso (L5) identity, penalized	the shrunk coefficient $\hat{\beta}_m^\lambda$: pulled toward 0, depends on λ and on the units of x_m; biased by design.	—	It is a coefficient for prediction and should not be read as the effect of x_m. A lasso estimate of zero means that x_m was not selected.

2 Predicted quantities

2.1 Odds ratios and changes in probability

In the class logit, $\hat{\beta}_{\mathrm{oil}} = -1.560$, so oil lowers the log odds of democracy by 1.56 for every country and multiplies the odds by $e^{\hat{\beta}_{\mathrm{oil}}} = 0.21$ for every country. The change in probability varies across countries (Figure 1, right; Table 2). At the three values of lgdp marked in Figure 1 it is -5, -35 and -17 percentage points. The left panel of Figure 1 shows the model on the log-odds scale, where oil shifts a straight line down by a constant amount. The right panel shows the probability scale, where the same shift is small in the tails of the curve and large in the middle.

The odds ratio is constant but the change in probability is not. In the middle row of Table 2, a country with $\hat{\pi} = 0.50$ has odds of $0.50/0.50 = 1.00$. Oil multiplies the odds by 0.21, so the new odds are 0.21 and the new probability is $0.21/1.21 = 0.17$, a decrease of 33 points. The same calculation gives a decrease of 8 points for a country starting at 0.10 and 25 points for a country starting at 0.90. The size of the change depends on the starting value of π_i, which depends on all of x_i (here, on lgdp). We do not observe π_i, because each country either is or is not a democracy, so it has to be estimated from the model. This is the reason for computing predicted quantities. The derivative gives the same result. $\partial\hat{\pi}(x)/\partial x_j = \hat{\beta}_j \hat{\pi}(x)\{1 - \hat{\pi}(x)\}$ is largest at $\hat{\pi} = 0.5$, where $\hat{\pi}(1-\hat{\pi}) = 0.25$ (for lgdp, $1.851 \times 0.25 = 0.46$ per unit), and approaches zero as $\hat{\pi}$ approaches 0 or 1. It follows that a logit coefficient divided by 4 is the largest possible change in probability per unit of x_j, and this maximum applies only to units near $\hat{\pi} = 0.5$.

2.2 Choosing what to report (L8)

A predicted quantity is a value implied by the model on the scale of the outcome, such as a probability, an expected count or a survival probability, computed from $\hat{\beta}$ at chosen values of x. Reporting one involves three choices, and each should be stated.

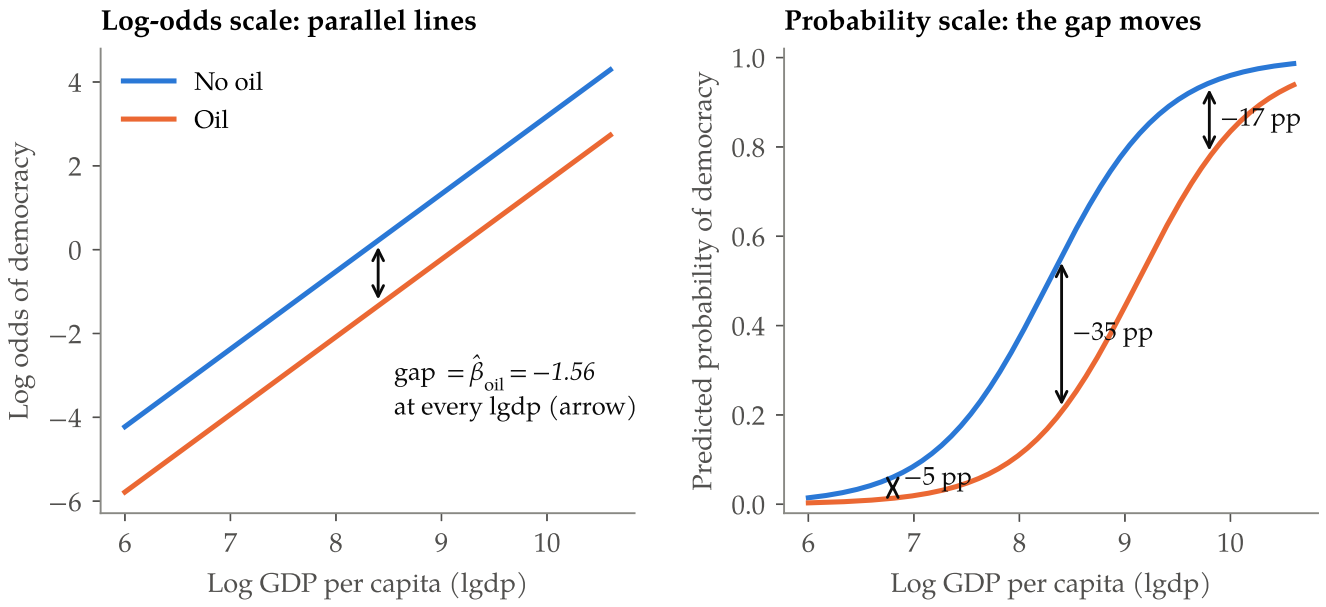


Figure 1: Logit of regime type on oil and lgdp, with the class estimates ($\hat{\beta}_0 = -15.329$, $\hat{\beta}_{\text{oil}} = -1.560$, $\hat{\beta}_{\text{lgdp}} = 1.851$; L8). The gap between the curves with and without oil is constant on the log-odds scale and varies on the probability scale.

Table 2: Switching oil on multiplies every country's odds by $e^{\hat{\beta}_{\text{oil}}} = 0.21$ (L8). By hand, new odds = old odds $\times 0.21$ and $\pi = \text{odds}/(1 + \text{odds})$.

$\hat{\pi}$ without oil	odds	odds with oil	$\hat{\pi}$ with oil	change (pp)
0.1	0.11	0.02	0.02	−8
0.5	1.00	0.21	0.17	−33
0.9	9.00	1.89	0.65	−25

Contrast. $\hat{\tau}_j(x) = \hat{\pi}(x_j = a, x_{-j}) - \hat{\pi}(x_j = b, x_{-j})$ is the difference between the predicted probability with $x_j = a$ and with $x_j = b$, other covariates unchanged. For a binary x_j, use $a = 1$ and $b = 0$. For a continuous x_j, choose a change that is easy to describe, such as one unit of lgdp (a 2.72-fold increase in GDP per capita), a doubling of GDP per capita ($c = \log 2$), one standard deviation, or the interquartile range. The derivative is the limit as the change goes to zero. Different contrasts give different numbers (21.7, 16.0, 40.4 and 49.7 points in L8), so the contrast should always be named.

Values of the other covariates. The contrast depends on the other covariates. The PEA (partial effect at the average) holds them at typical values, $\hat{\tau}^{\text{PEA}} = \hat{\pi}(1, \overline{\text{lgdp}}) - \hat{\pi}(0, \overline{\text{lgdp}})$ (L8: −0.366). It describes a single hypothetical country with average characteristics. The APE (average partial effect) computes the contrast for each country at its observed x_i and averages the results, $\hat{\tau}^{\text{APE}} = \frac{1}{n}\sum_i \hat{\tau}_i$ (−0.229). It describes the countries in the data. Because the model is nonlinear, the effect at the average country is generally different from the average effect. Hanmer and Kalkan (2013) recommend the observed-value approach (APE).

Uncertainty. The reported quantity $T(\hat{\beta})$ is a function of $\hat{\beta}$ and inherits its uncertainty. Since $\hat{\beta} \overset{\text{approx.}}{\sim} N\big(\beta, [-H(\hat{\beta})]^{-1}\big)$, draw $\beta^{(k)} \sim N\big(\hat{\beta}, \widehat{\text{Var}}(\hat{\beta})\big)$ for $k = 1, \ldots, R$, compute $T(\beta^{(k)})$, and report the standard deviation and the 2.5th and 97.5th percentiles of the draws. The delta method is an alternative (L8). Simulation is used because T is a nonlinear function of $\hat{\beta}$, so its standard error is not a simple multiple of $\text{se}(\hat{\beta})$. The spread of T across draws of β gives its uncertainty without further algebra.

Elements of a complete interpretation. The midterm rewards interpretations that state (i) the quantity and the contrast, (ii) the size of the change in x, (iii) the values at which the other covariates are held (PEA or APE), (iv) the scale, and (v) the uncertainty.

An example: "Holding lgdp at each country's observed value, switching oil from 0 to 1 lowers the predicted

probability of democracy by 23 percentage points on average (APE −0.229, 95% interval [−0.346, −0.111]; L8).” The quantity and contrast are $\hat{\pi}$ for oil equal to 1 versus 0, the change is switching oil on, the covariates are held at observed values, the scale is percentage points, and the uncertainty is given by the interval.

3 Exercises for Part I

Exercise 1 asks for interpretations in words, Exercises 2–7 ask for predicted quantities computed by hand, and Exercise 8 asks you to find errors in interpretation. The instructor version shows each step of the answers.

Exercise 1. Interpreting coefficients in words

Setup. Each row gives a coefficient from a model estimated in lecture, along with the outcome and covariates. Use the steps in §1: read the sign, exponentiate if the link is not the identity, and state what the multiplier applies to. A complete sentence states what is held fixed, the size of the change in x and the scale, and uses units that a reader understands, such as percentage points or “times as many”.
Task. Write one sentence for each row. For the log-scale rows, the values of $e^{0.5}$, e^{1} and e^{2} in the caption of Table 3 can be used to check the size of each multiplier.

Model, outcome and covariates	Output	One sentence
OLS (L2): turnout (pp) on age and mail	$\hat{\beta}_{\text{age}} = 0.585$	Among units with the same mail status, turnout is 0.585 points higher on average for each additional year of age (the slope after mail is partialled out of both variables).
LPM (L7): democracy (0/1) on oil and lgdp	$\hat{\beta}_{\text{oil}} = -0.22$	Holding lgdp fixed, oil states are 22 percentage points less likely to be democracies.
Logit (L8): democracy on oil and lgdp	$\hat{\beta}_{\text{oil}} = -1.560$	Holding lgdp fixed, oil lowers the log odds of democracy by 1.56, which multiplies the odds by $e^{-1.56} = 0.21$ (a 79% reduction). The change in probability depends on the country’s starting probability.
Ordered logit (L9): closed autocracy, electoral autocracy, democracy, on lgdp and oil	$\hat{\beta}_{\text{lgdp}} = 1.96$	A one-unit increase in lgdp raises the log odds of being above either cutpoint by 1.96, which multiplies those odds by $e^{1.96} = 7.1$. Doubling GDP per capita multiplies them by $2^{1.96} = 3.9$.
Multinomial logit (L9): military, single-party or personalist, on lgdp and oil; base category military	$\hat{\beta}_{\text{pers,oil}} = -2.37$	Oil lowers the log odds of personalist rule relative to military rule by 2.37, which multiplies $P(\text{pers})/P(\text{mil})$ by $e^{-2.37} = 0.09$. The coefficient alone does not show how $P(Y_i = \text{pers} \mid x_i)$ changes.
Poisson (L10): protests per year on lgdp, e_autocracy and oil	$\hat{\beta}_{\text{e_autocracy}} = 0.63$	Holding lgdp and oil fixed, electoral autocracies are expected to have $e^{0.63} = 1.88$ times as many protests (an incidence rate ratio, or 88% more).
Cloglog hazard (L12): yearly probability that a regime ends, on lgdp and oil	$\hat{\beta}_{\text{lgdp}} = -0.73$	In each spell year, a one-unit increase in lgdp multiplies the hazard of regime breakdown by $e^{-0.73} = 0.48$ (52% lower). Doubling GDP per capita multiplies it by 0.60.

For the identity-link models (OLS and LPM), the coefficient is the change in the mean and can be read in the units of the outcome. In the LPM the units are probability points, so −0.22 is 22 percentage points. The OLS sentence refers to units with the same mail status because, by the Frisch–Waugh–Lovell theorem, 0.585 is the slope of turnout on age after mail has been partialled out of both.

For the models with log-odds links, exponentiate and state what the multiplier applies to. In the logit it applies to the odds of democracy. In the ordered logit the same multiplier applies at both cutpoints (proportional odds), so one sentence covers the odds of being above closed autocracy and the odds of being above electoral autocracy. In the multinomial logit it applies to the ratio of two probabilities, personalist to military, and not to $P(\text{pers})$ itself (see Exercise 5).

In the Poisson model $e^{0.63} = 1.88$ is a ratio of expected counts, and the difference in the number of protests depends on the baseline count (Exercise 6). In the cloglog model $e^{-0.73} = 0.48$ multiplies the yearly hazard, a 52% reduction, and the multiplier for a doubling is $2^{-0.73} = 0.60$.

A frequent mistake is to report the coefficient itself as a change in probability or in the count. This is correct only for the two identity-link models.

Exercise 2. The class logit (L8)

Setup. The class logit of democracy ($Y_i = 1$) on oil and lgdp has $\hat{\beta}_0 = -15.329$, $\hat{\beta}_{\text{oil}} = -1.560$ and $\hat{\beta}_{\text{lgdp}} = 1.851$. A country's linear predictor is $\hat{\eta} = -15.329 - 1.560\,\text{oil} + 1.851\,\text{lgdp}$, its predicted odds are $e^{\hat{\eta}}$, and its predicted probability is $\hat{\pi} = e^{\hat{\eta}}/(1+e^{\hat{\eta}}) = 1/(1+e^{-\hat{\eta}})$. Doubling income adds $\log 2 = 0.69$ to lgdp.
Task. (a) Interpret $\hat{\beta}_{\text{oil}}$ on the log-odds scale and as an odds ratio. Why is neither enough for a reader? (b) By what factor does doubling GDP per capita multiply the odds? (c) For a country with lgdp $= 8.5$, compute $\hat{\pi}$ with and without oil, and the change. (d) The slides report $\hat{\tau}^{\text{PEA}} = -0.366$ and $\hat{\tau}^{\text{APE}} = -0.229$. Why do they differ, and which would you report?

(a) On the log-odds scale, among countries with the same lgdp, oil producers have log odds of democracy that are 1.56 lower. As an odds ratio, $e^{-1.56} = 0.21$, so oil multiplies the odds of democracy by 0.21, a 79% reduction. Equivalently, the odds for non-oil countries are 4.8 times as large. Neither statement describes a change in probability, and the same odds ratio corresponds to different changes in probability depending on the starting point. In Table 2 the odds ratio of 0.21 produces decreases of 8, 33 and 25 points.

(b) Doubling is a change of $c = \log 2$ in lgdp, so the multiplier is $e^{c\hat{\beta}} = e^{0.69\times 1.851} = 2^{1.851} = 3.6$. Holding oil fixed, doubling GDP per capita multiplies the odds of democracy by about 3.6. This differs from 2×1.851 and from $e^{1.851} = 6.37$, which is the multiplier for a 2.72-fold increase.

(c) Compute $\hat{\eta}$, then the odds, then $\hat{\pi}$. Without oil, $\hat{\eta} = -15.329 + 1.851 \times 8.5 = 0.4045$, the odds are $e^{\hat{\eta}} = 1.50$ and $\hat{\pi} = 1.50/(1+1.50) = 0.60$. With oil, $\hat{\eta} = 0.4045 - 1.560 = -1.1555$, the odds are 0.31 (equal to 1.50×0.21), and $\hat{\pi} = 1/(1+e^{-\hat{\eta}}) = 1/(1+3.18) = 0.24$. The change is $0.24 - 0.60$, or -36 percentage points. In words, for a country with lgdp of 8.5, oil lowers the predicted probability of democracy from 0.60 to 0.24, a decrease of 36 points. The change is neither -1.56 points nor -156.

(d) The PEA computes the contrast for one country with lgdp at its mean, switching oil from 0 to 1. The APE computes the contrast for each country at its observed lgdp and averages the results. The logistic curve is steep in the middle and flat in the tails. The country with mean lgdp lies near the steep part, where the effect is large, while many observed countries lie in the tails, where oil changes the probability very little (Figure 1, right: 5 and 17 points in the tails compared with 35 near the centre). Averaging over the observed countries therefore gives a smaller effect. Report the APE, which describes the countries in the data (Hanmer and Kalkan), with its interval: -0.229, 95% interval $[-0.346, -0.111]$ (L8).

Exercise 3. Logit predictions by hand

Setup. A hypothetical logit of turnout ($Y_i = 1$ if person i votes) has $\hat{\eta} = -3 + 1 \cdot$ decades since age $18 + 2 \cdot$ contact, where contact equals 1 if the person received a mobilization call and 0 otherwise. Ages 18, 48 and 68 correspond to 0, 3 and 5 decades since age 18. All values of $\hat{\eta}$ here are integers, so each $\hat{\pi}$ can be read from Table 3.
Task. (a) What is the odds ratio for contact? (b) Find $\hat{\pi}$ with and without contact at ages 18, 48 and 68, and the change at each age. (c) What single number would you report for the effect of contact, and what other information is needed?

Table 3: Pocket table: $\text{logit}^{-1}(z) = e^z/(1+e^z)$. Also useful: $e^{0.5} = 1.65$, $e^1 = 2.72$, $e^2 = 7.39$, $e^{-1} = 0.37$, $e^{-2} = 0.14$, $\log 2 = 0.69$.

z	-4	-3	-2	-1	0	1	2	3	4
$\text{logit}^{-1}(z)$	0.018	0.047	0.119	0.269	0.500	0.731	0.881	0.953	0.982

(a) $e^2 = 7.39$. A call multiplies the odds of voting by about 7.4 at every age, because the model has no interaction between age and contact.

(b) Without contact $\hat{\eta} = -3 +$ decades, and contact adds 2. The probabilities from Table 3 are as follows. Age 18: $\hat{\eta} = -3 \to -1$, $\hat{\pi} = 0.05 \to 0.27$ (+0.22). Age 48: $\hat{\eta} = 0 \to 2$, $\hat{\pi} = 0.50 \to 0.88$ (+0.38). Age 68: $\hat{\eta} = 2 \to 4$, $\hat{\pi} = 0.88 \to 0.98$ (+0.10). The odds ratio is the same at each age but the change in probability is not. The change is largest at age 48, where $\hat{\pi}$ starts at 0.5 and the curve is steepest. At age 18 the starting probability is low enough that the call still leaves voting unlikely. At age 68 the probability of voting is already 0.88, which leaves little room for an increase.

(c) Report the APE of contact. Compute the contrast between contact and no contact at each person's observed age, average over the sample, and report a simulation or delta-method interval. State the contrast and the scale (percentage points). The odds ratio of 7.4 can be reported as well, but a reader cannot convert it into a change in turnout without the baseline probability.

Exercise 4. Ordered logit (L9)

Setup. The outcome has three ordered categories: closed autocracy (1), electoral autocracy (2) and democracy (3). The model assumes a latent variable $Y_i^* = x_i^\top \beta + \varepsilon_i$ with a logistic error. A country is a closed autocracy if Y_i^* is below the first cutpoint ζ_1, an electoral autocracy if it is between ζ_1 and ζ_2, and a democracy if it is above ζ_2. The cutpoints replace the intercept. The class estimates are $\hat{\beta}_{\text{lgdp}} = 1.96$, $\hat{\beta}_{\text{oil}} = -1.54$, $\hat{\zeta}_1 = 13.75$ and $\hat{\zeta}_2 = 16.30$.

Formulas. $P(\text{closed}) = \text{logit}^{-1}(\hat{\zeta}_1 - x_i^\top \hat{\beta})$ and $P(\text{closed or electoral}) = \text{logit}^{-1}(\hat{\zeta}_2 - x_i^\top \hat{\beta})$. $P(\text{electoral})$ is the difference between these two, and $P(\text{democracy}) = 1 - \text{logit}^{-1}(\hat{\zeta}_2 - x_i^\top \hat{\beta})$.

Task. Complete the table for countries with lgdp of 8.5 and 6.0, with and without oil. Then answer the following. (a) Which categories change in the direction implied by the sign of $\hat{\beta}_{\text{oil}}$? (b) Why do the three changes sum to zero? (c) At approximately what value of lgdp does the effect of oil on electoral autocracy change sign?

lgdp	oil	$x_i^\top \hat{\beta}$	$\hat{\zeta}_1 - x_i^\top \hat{\beta}$	$\hat{\zeta}_2 - x_i^\top \hat{\beta}$	$\hat{P}$(closed)	$\hat{P}$(electoral)	$\hat{P}$(democracy)
8.5	0	16.66	−2.91	−0.36	0.05	0.36	0.59
8.5	1	15.12	−1.37	1.18	0.20	0.56	0.24
change from switching oil on					+0.15	+0.20	−0.35
6.0	0	11.76	1.99	4.54	0.88	0.11	0.01
6.0	1	10.22	3.53	6.08	0.97	0.03	0.00
change from switching oil on					+0.09	−0.08	−0.01

For lgdp of 8.5 without oil, $x_i^\top \hat{\beta} = 1.96 \times 8.5 = 16.66$. Then $\hat{\zeta}_1 - x_i^\top \hat{\beta} = -2.91$, so $P(\text{closed}) = \text{logit}^{-1}(-2.91) = 0.05$, and $\hat{\zeta}_2 - x_i^\top \hat{\beta} = -0.36$, so $P(\text{closed or electoral}) = \text{logit}^{-1}(-0.36) = 0.41$. This gives $P(\text{electoral}) = 0.41 - 0.05 = 0.36$ and $P(\text{democracy}) = 1 - 0.41 = 0.59$. Oil subtracts 1.54 from $x_i^\top \hat{\beta}$, so both differences increase by 1.54. The cumulative probabilities rise and probability shifts toward the lower categories.

(a) Only the two end categories must move in the direction of the sign. A negative $\hat{\beta}_{\text{oil}}$ shifts the latent distribution to the left, so $P(\text{closed})$ rises and $P(\text{democracy})$ falls at both values of lgdp. The middle category gains countries from democracy and loses countries to closed autocracy, and the net change depends on where the country is located. At lgdp of 8.5 most of the probability is in democracy and electoral autocracy, so oil mainly moves countries from democracy to electoral autocracy (+0.20). At lgdp of 6.0 most of the probability is in closed and electoral autocracy, so oil mainly moves countries from electoral to closed autocracy (−0.08). In the formula $P(\text{electoral}) = \text{logit}^{-1}(\zeta_2 - x_i^\top \beta) - \text{logit}^{-1}(\zeta_1 - x_i^\top \beta)$, $x_i^\top \beta$ appears in both terms and both terms move in the same direction, so the sign of β does not determine the sign of the difference.

(b) The three probabilities sum to one with and without oil, so any increase in one category is offset by decreases in the others: $0.15 + 0.20 - 0.35 = 0$ at lgdp of 8.5 and $0.09 - 0.08 - 0.01 = 0$ at lgdp of 6.0.

(c) The change in $P(\text{electoral})$ is positive at 8.5 and negative at 6.0, so it equals zero somewhere in between, at lgdp of about 8.1 (the dashed line in the L9 figure). At that point the inflow from democracy equals the outflow to closed autocracy.

Exercise 5. Multinomial logit (L9)

Setup. The outcome is the type of autocracy, with three unordered categories: military (the base category), single-party and personalist. Each non-base category j has its own coefficient vector β_j. For the base category $\beta_1 = 0$, so its score is always $e^0 = 1$. The probability of a category is its score divided by the sum of the scores, $P(Y_i = j \mid x_i) = \exp(x_i^\top \beta_j) / \sum_k \exp(x_i^\top \beta_k)$. The class estimates for (intercept, lgdp, oil) are $\hat{\beta}_{\text{sp}} = (-12.78,\ 1.75,\ -0.94)$ for single-party and $\hat{\beta}_{\text{pers}} = (-25.68,\ 3.34,\ -2.37)$ for personalist.

Task. For lgdp of 8, compute the scores $\exp(x_i^\top \hat{\beta}_j)$ and the probabilities with and without oil. Then use these numbers to explain why oil raises the probability of single-party rule although its single-party coefficient is negative.

	score $\exp(x_i^\top \hat{\beta}_j)$			probability		
	military	single-party	personalist	military	single-party	personalist
oil = 0	1.00	3.39	2.83	0.14	0.47	0.39
oil = 1	1.00	1.32	0.26	0.39	0.51	0.10
change				+0.25	+0.04	−0.29

Without oil, the linear predictors are $-12.78 + 1.75 \times 8 = 1.22$ for single-party, $-25.68 + 3.34 \times 8 = 1.04$ for personalist and 0 for military. With oil, each category's own oil coefficient is added, giving $1.22 - 0.94 = 0.28$ for single-party and $1.04 - 2.37 = -1.33$ for personalist, while military stays at 0. The scores without oil are $e^{1.22} = 3.39$, $e^{1.04} = 2.83$ and 1, with a total of 7.22. With oil they are $e^{0.28} = 1.32$, $e^{-1.33} = 0.26$ and 1, with a total of 2.59. Dividing each score by the total gives probabilities of $(0.14, 0.47, 0.39)$ without oil and $(0.39, 0.51, 0.10)$ with oil, for military, single-party and personalist.

The probability of a category is its share of the total score. Oil reduces the single-party score from 3.39 to 1.32, by a factor of $e^{-0.94} = 0.39$. It reduces the personalist score from 2.83 to 0.26, by a factor of 0.09, and leaves the military score at 1. The total falls from 7.22 to 2.59, by a factor of 0.36, which is a larger proportional decrease than for the single-party score. As a result the single-party share rises from 0.47 to 0.51. The coefficient of -0.94 compares single-party with the base category only. The ratio $P(\text{sp})/P(\text{mil})$ falls from 3.39 to 1.32, mostly because the probability of military rule rises from 0.14 to 0.39. The three changes sum to zero $(0.25 + 0.04 - 0.29 = 0)$.

In a multinomial logit, the sign of $\hat{\beta}_{jm}$ describes the change in $P(j)/P(\text{base})$. The change in $P(j)$ itself has to be computed from predicted probabilities or the APE.

Exercise 6. Poisson regression (L10)

Setup. The outcome is the number of protests in a country per year. The Poisson model uses a log link, so the coefficients act on the log of the expected count μ_i. The class estimates are $\log \hat{\mu}_i = -2.10 + 0.30\,\text{lgdp}_i + 0.63\,\text{e_autocracy}_i - 0.03\,\text{oil}_i$, where e_autocracy equals 1 for electoral autocracies and 0 otherwise. A student writes: "electoral autocracies have 0.63 more protests."

Task. (a) What is wrong with this statement? (b) Give the correct interpretation. (c) For oil $= 0$, compute $\hat{\mu}$ for e_autocracy $= 0$ and 1 at lgdp $= 8$ and lgdp $= 10$, and the difference in each case. (d) What would you report if asked how many more protests electoral autocracies have?

(a) 0.63 is a change in $\log \mu$. Adding 0.63 to $\log \mu$ multiplies μ by $e^{0.63}$. It does not add 0.63 protests.

(b) Holding lgdp and oil fixed, electoral autocracies are expected to have $e^{0.63} = 1.88$ times as many protests. This is an incidence rate ratio and corresponds to 88% more protests.

(c) At lgdp $= 8$, $\hat{\eta} = -2.10 + 0.30 \times 8 = 0.30$ and $\hat{\mu} = e^{0.30} = 1.35$. With e_autocracy $= 1$, $\hat{\mu} = e^{0.93} = 2.53$. The difference is 1.18. At lgdp $= 10$, $\hat{\mu}$ is $e^{0.90} = 2.46$ without and $e^{1.53} = 4.62$ with, a difference of 2.16. The ratio is 1.88 in both cases, but the difference changes (Figure 2, right). The difference equals $\hat{\mu}_0(1.88 - 1)$, so it increases with the baseline count. Because the lgdp coefficient is positive, richer countries have a higher baseline count and therefore a larger difference.

(d) Report the APE, about 1.5 more protests per year on average (1.54, se 0.31; L10). The APE averages the difference over the countries in the data and is measured in protests. The IRR of 1.88 can be reported alongside it.

Exercise 7. Discrete-time hazard model (L12)

Setup. Each regime is observed year by year. The hazard h_{ij} is the probability that regime i ends in spell year j, given that it lasted until year j. The model uses a complementary log-log (cloglog) link, $\log\{-\log(1 - h_{ij})\} = \gamma_j + x_i^\top \beta$, where γ_j is the baseline hazard for spell year j. Survival to year t is the probability that the regime does not end in any year up to t, $S(t \mid x) = \prod_{j \le t}(1 - h_{ij})$. The class estimates are $\hat{\beta}_{\text{lgdp}} = -0.73$ and $\hat{\beta}_{\text{oil}} = -0.53$.

Formulas. From L12, $1 - h_{ij} = e^{-\lambda_{ij}}$, where $\lambda_{ij} = e^{\gamma_j + x_i^\top \beta}$ is the hazard rate.

Task. (a) Interpret each coefficient as a hazard ratio. (b) A regime without oil has $\hat{S}(20 \mid x) = 0.50$. What is $\hat{S}(20 \mid x)$ for an otherwise identical regime with oil? (c) Why is the answer not 0.50×0.59 or $0.50/0.59$?

(a) A one-unit increase in lgdp multiplies the yearly hazard of regime breakdown by $e^{-0.73} = 0.48$, a 52% reduction. Oil multiplies it by $e^{-0.53} = 0.59$, a 41% reduction. Under proportional hazards both ratios apply in every spell year. A lower hazard means the regime is expected to last longer.

(b) $S(t \mid x) = \prod_{j \le t}(1 - h_{ij}) = \prod_{j \le t} e^{-\lambda_{ij}} = \exp\left(-\sum_{j \le t} \lambda_{ij}\right)$. Oil multiplies each λ_{ij} by $e^{\beta_{\text{oil}}}$, so it multiplies their sum by the same factor, and $S(t \mid x_{\text{oil}}) = \exp\left(-e^{\beta_{\text{oil}}} \sum_j \lambda_{ij}\right) = S(t \mid x_{\text{no oil}})^{e^{\beta_{\text{oil}}}}$. With the numbers, $0.50^{0.59} = 0.66$. A regime with a 50% chance of lasting 20 years without oil has a 66% chance with oil, an increase of 16 points.

(c) $0.50 \times 0.59 = 0.30$ would imply that oil reduces survival, which contradicts the lower hazard. $0.50/0.59 = 0.85$ treats the hazard ratio as a ratio of survival probabilities and is too large. The hazard ratio multiplies a rate that accumulates over years, and survival is the exponential of minus the accumulated rate. The hazard ratio therefore enters as an exponent on S.

Exercise 8. Errors in interpretation

Setup. Each sentence below could appear in an exam answer and contains one error, usually about the scale (log odds, odds or probability) or about the comparison (relative to the base category, or to the hazard rather than survival).
Task. Explain the error and write a corrected sentence.

	Sentence	Error and correction
1	"Oil lowers the probability of democracy by 1.56." (logit)	$\hat{\beta}$ is a change in log odds, and a probability cannot fall by more than 1. Correct version: oil multiplies the odds of democracy by 0.21, which lowers the probability by 36 points at lgdp of 8.5 and by 23 points on average (APE -0.229).
2	"Oil states are 0.21 times as likely to be democracies."	0.21 multiplies the odds, not the probability. At $\hat{\pi} = 0.5$ the probability falls to 0.17, so oil states are 0.34 times as likely to be democracies. The odds ratio approximates the ratio of probabilities only when π is close to 0.
3	"Electoral autocracies have 1.88 more protests."	1.88 is a ratio, so electoral autocracies have 1.88 times as many protests (88% more). The difference in protests is $\hat{\mu}(1.88 - 1)$, which is 1.18 at lgdp of 8 and 2.16 at lgdp of 10 (Exercise 6).
4	"Oil's coefficient on single-party is -0.94, so oil lowers $P(\text{single-party})$." (multinomial)	The coefficient compares single-party rule with the base category, military rule, so oil lowers $P(\text{sp})/P(\text{mil})$. $P(\text{single-party})$ itself rises from 0.47 to 0.51 because the total score falls by more (Exercise 5).
5	"Oil's coefficient is negative, so oil lowers every higher category." (ordered)	Only the first and last categories must move in the direction of the sign. $P(\text{electoral})$ rises at lgdp of 8.5 ($+0.20$) and falls at 6.0 (-0.08) (Exercise 4).
6	"A hazard ratio of 0.59 means oil regimes are 0.59 times as likely to survive 20 years."	The hazard ratio multiplies the yearly hazard of regime breakdown. Survival is raised to the power 0.59, from 0.50 to 0.66, so oil regimes are more likely to survive 20 years (Exercise 7).
7	"The logit coefficient (1.851) exceeds the probit's, so logit finds a bigger effect."	Logit and probit coefficients are on different scales (L7). The logistic error has a standard deviation of about 1.8 and the probit error a standard deviation of 1, so logit coefficients are usually 1.6 to 1.8 times as large for the same data. Compare $\hat{\pi}$ or the APE, which are nearly the same in the two models.
8	"The lasso set $\hat{\beta}_m = 0$, so x_m has no effect."	A zero means that the lasso did not select x_m at this value of λ, often because a correlated covariate carries similar information. Penalized coefficients are biased toward zero by design and are meant for prediction, not for estimating effects.

Part II Maximum likelihood versus Bayesian inference

4 The likelihood

Both approaches start from a probability model for the data and from the data themselves. Here the model assumes that each respondent independently says yes with the same probability, $Y_i \sim \text{Bernoulli}(\pi)$. The likelihood (L6) is the probability of the observed data, written as a function of π,

$$L(\pi; Y) = p(Y \mid \pi) = \prod_{i=1}^{n} \pi^{Y_i}(1-\pi)^{1-Y_i} = \pi^r(1-\pi)^{n-r}, \qquad \ell(\pi) = r\log\pi + (n-r)\log(1-\pi),$$

where $r = \sum_i Y_i$ is the number of yes answers. In the running example (L13), $n = 15$ respondents in Hainan, $r = 12$ of them trust local government, and π is the share of all adults in Hainan who do. In both approaches π is fixed and unknown (L13). The approaches differ in what they treat as random and therefore in how they define uncertainty. Table 4 compares them using the same data.

ML asks which value of π makes the observed data most likely and then asks how much this estimate would vary across repeated surveys. Bayesian inference asks how plausible each value of π is, given the prior and the observed data. The ML standard error comes from the curvature of the log likelihood at its maximum, $-H(\hat{\pi}) = r/\hat{\pi}^2 + (n-r)/(1-\hat{\pi})^2 = n/\{\hat{\pi}(1-\hat{\pi})\} = 15/(0.80 \times 0.20) = 93.75$, so $\text{se} = 1/\sqrt{93.75} = 0.103$. A more sharply curved log likelihood gives a smaller standard error. For Hainan, the Wald interval $[0.598, 1.002]$ extends past 1 because the normal approximation is poor with $n = 15$ and $\hat{\pi}$ close to 1. The posterior stays within $[0, 1]$ by construction. This is a limitation of the normal approximation and not of ML in general.

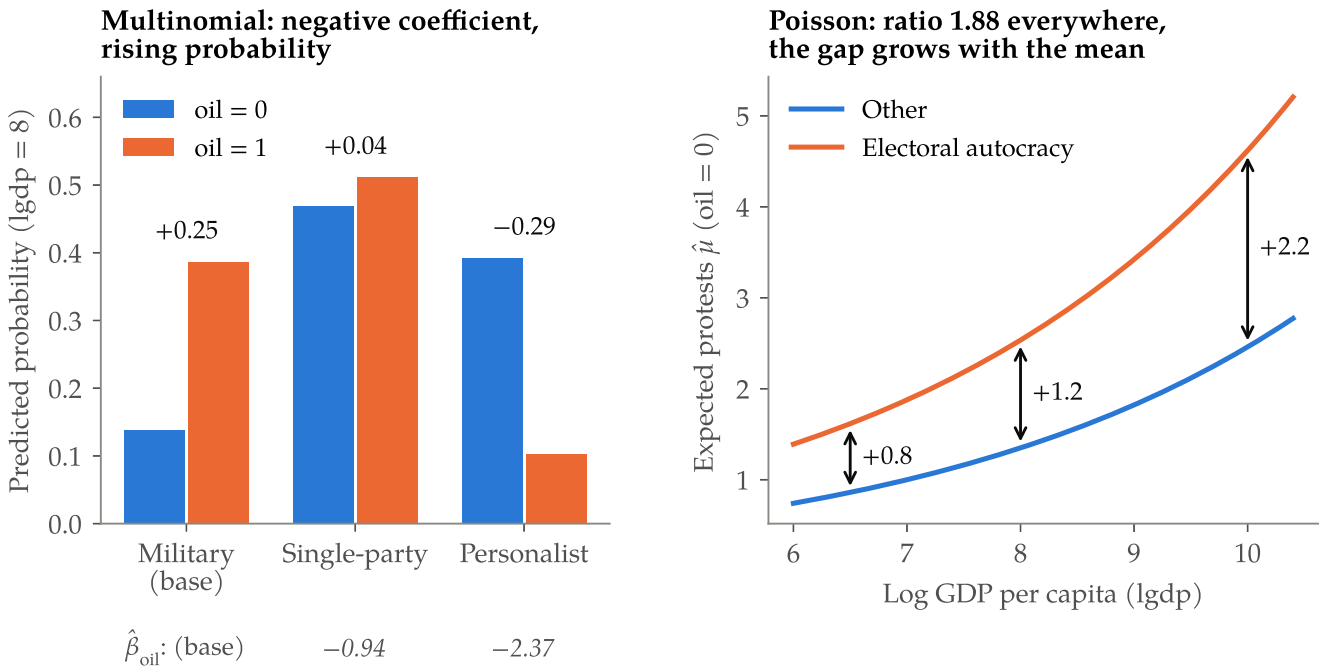


Figure 2: Left (Exercise 5): multinomial probabilities at lgdp = 8, military as base. Right (Exercise 6): Poisson expected protests at oil = 0, from the two-decimal coefficients printed on the L10 slide (the slide's own figure uses unrounded estimates, so its gaps differ slightly).

5 Prior, likelihood and posterior

The prior is $\pi \sim \text{Beta}(a, b)$, with $\text{E}[\pi] = a/(a+b)$. The Beta distribution is used because it is defined on $[0, 1]$, like π, and its parameters can be read as prior counts of yes and no answers. Matching the sample shares in the other 30 provinces (mean 0.58, sd 0.15) gives $\text{Beta}(6, 4)$, with mean 0.60, sd 0.15 and a "worth" of $a + b = 10$ respondents. The likelihood is $L(\pi; Y) = \pi^{12}(1-\pi)^3$. Bayes' rule combines the two:

$$\underbrace{p(\pi \mid Y)}_{\text{posterior}} = \frac{\overbrace{p(Y \mid \pi)}^{\text{likelihood } L(\pi;Y)} \overbrace{p(\pi)}^{\text{prior}}}{\underbrace{p(Y)}_{\text{normalizing constant}}} \propto L(\pi; Y)\, p(\pi).$$

The term $p(Y)$ does not depend on π and only makes the posterior integrate to one, so it can be dropped when we are interested in the shape of the posterior. With a Bernoulli likelihood the Beta prior is conjugate, which means that the posterior is also a Beta distribution. Multiplying $\pi^{a-1}(1-\pi)^{b-1}$ by $\pi^r(1-\pi)^{n-r}$ adds the exponents, so

$$\pi \mid Y \sim \text{Beta}(a + r,\ b + n - r) = \text{Beta}(18, 7),$$

$$\text{E}[\pi \mid Y] = \underbrace{\tfrac{a+b}{a+b+n}}_{\text{weight on prior}} \cdot \tfrac{a}{a+b} + \underbrace{\tfrac{n}{a+b+n}}_{\text{weight on data}} \cdot \tfrac{r}{n} = 0.40 \cdot 0.60 + 0.60 \cdot 0.80 = 0.72.$$

The posterior mean is a weighted average of the prior mean and the MLE. The weights depend only on the prior worth, $a + b = 10$, and the sample size, $n = 15$, so the data receive $15/25 = 60\%$ of the weight. As n increases, the weight on the data approaches 100% and the prior has less influence.

The posterior lies between the prior and the likelihood, closer to whichever carries more information. For Hainan the prior receives 40% of the weight and moves the estimate by up to 11 points (under the strongest prior in Table 5). For Guangdong it receives 4% of the weight and moves the estimate by at most 1 point. In Figure 3 the Hainan posterior lies clearly between the prior and the likelihood, while the Guangdong posterior nearly coincides with the likelihood. The prior has the most influence when the sample is small, which is also when its source most needs to be justified.

Table 4: The two approaches on the same data (Hainan, $r = 12$, $n = 15$). Numbers from L6, L8 and L13.

	Maximum likelihood	Bayesian inference
Ingredients	model for Y only	model for Y *and* a prior $p(\pi)$
The likelihood	$L(\pi; Y)$: a function of π, not a distribution for it	the same $L(\pi; Y)$, now multiplied by the prior
Target	the π that maximizes ℓ: $\hat{\pi} = r/n = 0.80$	a distribution, the posterior $p(\pi \mid Y) \propto L(\pi; Y)\, p(\pi)$
What is random	the data, over repeated samples, so $\hat{\pi}$ varies from sample to sample	π is fixed, and $p(\pi \mid Y)$ describes uncertainty about it given the observed data
Uncertainty	$\text{se}(\hat{\pi}) = \sqrt{\hat{\pi}(1-\hat{\pi})/n} = 0.103$, from the curvature $-H(\hat{\pi}) = 93.75$ (observed information)	the spread of the posterior: sd $= 0.088$
Interval	95% **confidence** interval $\hat{\pi} \pm 1.96\,\text{se} = [0.598, 1.002]$. The 95% describes the procedure over repeated samples	95% **credible** interval $[0.53, 0.87]$: given our model and prior, a 95% probability that π lies in it
Probability of a claim	none, since π is not random and "$P(\pi > 0.6)$" has no meaning	$P(\pi > 0.6 \mid Y) = 0.90$
Computation	solve the score $s(\hat{\theta}) = 0$: closed form, Newton–Raphson (Fisher scoring in `glm()`), `optim()`	closed form if the prior is conjugate; otherwise grid, Metropolis/MCMC, Stan (HMC)
Assumes	the model is true; a random sample; n large for $\hat{\beta} \overset{\text{approx.}}{\sim} N$	the model and the prior, an additional assumption that is stated explicitly (L13)

Table 5: Posterior means under four priors (L13). "Worth" is $a + b$. A prior with a larger worth is more concentrated and carries more information.

Prior	Worth	Prior sd	Hainan ($n = 15$)	Guangdong ($n = 236$)
None (MLE)			0.800	0.466
Beta$(1, 1)$	2	0.29	0.765	0.466
Beta$(6, 4)$	10	0.15	0.720	0.472
Beta$(12, 8)$	20	0.11	0.686	0.477

Summary of the components. The prior $p(\pi)$ describes which values of π are considered plausible before seeing Y. It is an assumption, which can be summarized as pseudo-data worth $a + b$ respondents, and it does not imply that π is random. The likelihood $L(\pi; Y)$ measures how well each value of π accounts for the observed data. It is not a probability distribution for π, since its integral is not one, and it is the only contribution of the data in both approaches. The posterior $p(\pi \mid Y)$ is the product of prior and likelihood, rescaled to integrate to one, and describes uncertainty about π after observing the data. It can be summarized by its mean, a credible interval or a probability such as $P(\pi > 0.6 \mid Y)$. Under ML, the uncertainty of an estimate is the sampling variability of $\hat{\pi}$ (standard error and confidence interval). Under Bayesian inference it is the spread of the posterior (standard deviation and credible interval).

6 Sampling variability and posterior uncertainty

Figure 4 compares the two. The left panel shows the ML view. With π fixed at 0.67 (the value used in the L13 simulation), 20,000 surveys of $n = 15$ are drawn and $\hat{\pi}$ is computed for each. The standard deviation of these estimates is about 0.12, as in the L13 table, and this sampling variability is the basis of a confidence interval. The observed $\hat{\pi} = 0.80$ is one draw from this distribution, and a different sample of 15 respondents would give a different estimate. The right panel shows the Bayesian view. It uses only the observed survey and describes uncertainty about the fixed π with the posterior, without reference to repeated surveys.

Correct statements. For the confidence interval: "If the survey were repeated many times, 95% of the intervals constructed in this way would contain π." For the credible interval: "Given the model and the prior, the probability that π lies in $[0.53, 0.87]$ is 0.95."

Incorrect statement. For a single confidence interval: "The probability that π lies in $[0.598, 1.002]$ is 0.95." Once the interval is computed, it either contains the fixed π or it does not. The 95% refers to the procedure and not to a particular interval.

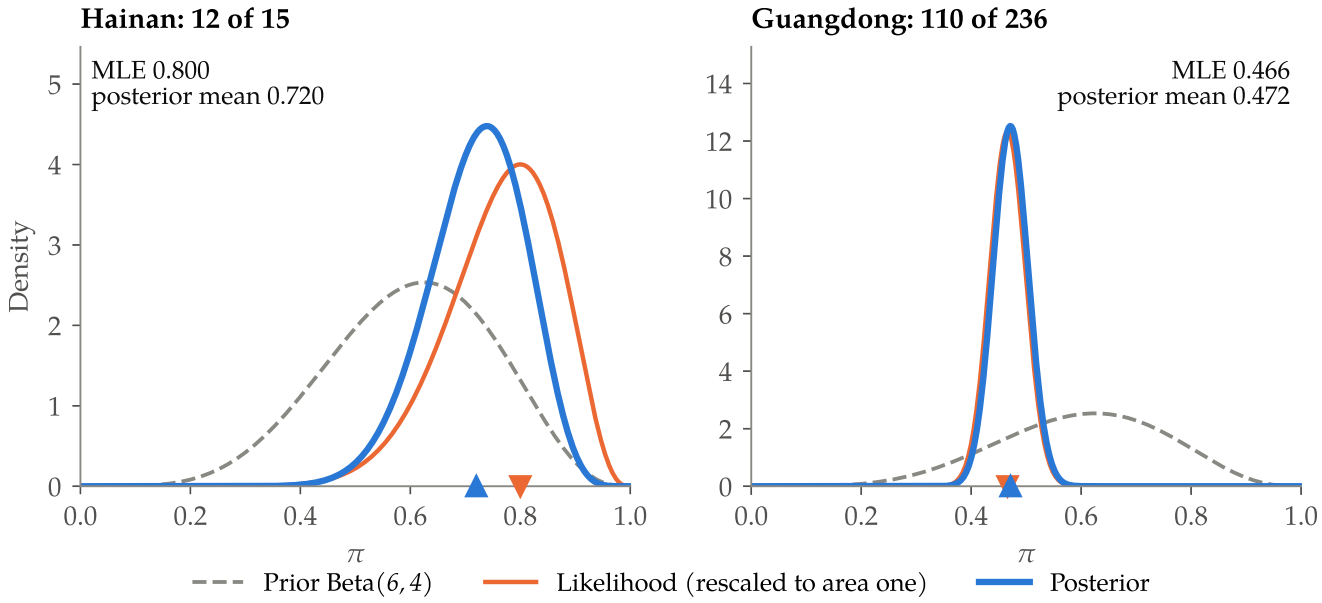


Figure 3: Prior, rescaled likelihood and posterior for two provinces under the same prior, Beta$(6,4)$. Triangles mark the MLE (orange, pointing down) and the posterior mean (blue, pointing up). The prior moves the estimate by 8 points for Hainan ($n = 15$) and by less than one point for Guangdong ($n = 236$).

7 Priors on regression coefficients

In regression models the priors are placed on the coefficients, for example $\beta_j \sim N(m_j, s_j^2)$. A normal prior is used instead of a Beta prior because coefficients can take any real value (L13). The prior mean m_j is the expected value of the coefficient, and the prior standard deviation s_j reflects how confident we are in it.

One reason to use a prior is separation. In the Hainan logit only one respondent is aged 60 or older, and she trusts local government. Increasing β_{60+} moves her predicted probability closer to 1 and always increases the likelihood, so the likelihood has no maximum. The MLE is $+\infty$. The `glm()` function stops at 16 with a standard error of 3,956 and predicts Pr(trust) $= 1.00$ based on one respondent. With a prior that gives little weight to extreme values, the product $L \times p$ has a finite maximum, because the likelihood levels off as β_{60+} increases while the prior density continues to fall.

Prior on β_{60+}	$\hat{\beta}_{60+}$ (truth 0.55)	Pr(trust), urban, 60+
None (`glm()`)	16.0 (se 3,956)	1.00
Weak, $N(0, 2.5^2)$	0.95 [−3.00, 5.43]	0.84 [0.24, 1.00]
From the other provinces, $N(0.55, 0.5^2)$	0.64 [−0.32, 1.65]	0.75 [0.45, 0.93]

Posterior probabilities depend on the prior. L13 lists four ways to limit or justify its influence. The first is to have enough data that the prior receives little weight, as in Guangdong. The second is to check sensitivity by reporting results under several priors. The third is to use weak priors. A $N(0, 2.5^2)$ prior on a logit coefficient excludes only odds ratios above about $e^5 \approx 148$. The fourth is to estimate the prior with a hierarchical model, $\beta_{0j} = \mu + u_j$, $u_j \sim N(0, \sigma_u^2)$, which estimates from the data how similar the provinces are (empirical Bayes and partial pooling; L14). In such a model a small province such as Hainan is shrunk toward the national average by an amount estimated from the data rather than chosen by the analyst.

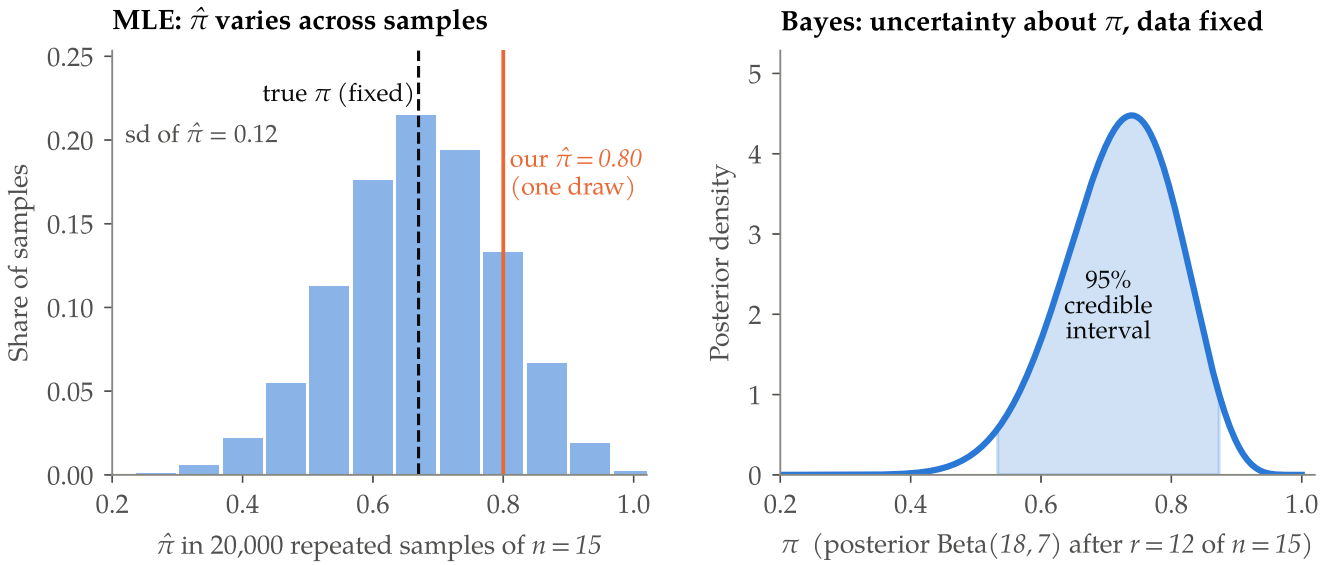


Figure 4: Left: sampling distribution of $\hat{\pi}$ across repeated surveys of $n = 15$ with $\pi = 0.67$. Right: posterior Beta$(18, 7)$ with its 95% credible interval.

> **Connection to Part I.** In a Bayesian regression with a weak prior, β_m has the same interpretation as before. It is a change in log odds, and e^{β_m} is an odds ratio. The difference is that β now has a posterior distribution. The posterior draws $\beta^{(k)}$ take the place of the draws from $N(\hat{\beta}, \widehat{\text{Var}}(\hat{\beta}))$ in the L8 simulation method. For each draw, compute $T(\beta^{(k)})$, for example a predicted probability or an APE, and summarize the results. Part I covers what an estimate means and Part II covers how to describe its uncertainty. In both cases the reported quantity can be computed from draws of β.

8 Approximating the posterior

Conjugate priors are the exception. For the logit no standard prior is conjugate, and the normalizing constant $p(Y) = \int L(\theta; Y)\, p(\theta)\, d\theta$, an integral over all values of all coefficients, has no closed form. The posterior therefore has to be approximated.

A grid approximation evaluates prior times likelihood at many values and rescales the results to sum to one (Exercise 10 uses five values). The number of evaluations grows quickly with the number of parameters. One coefficient at 100 values requires 100 evaluations, and four coefficients require 10^8.

The Metropolis algorithm (L13) instead generates a sequence of values from the posterior.

1. From the current value $\pi^{(t)}$, propose $\pi^* = \pi^{(t)} + \varepsilon$ with $\varepsilon \sim N(0, s^2)$.
2. Compute ratio $= \dfrac{L(\pi^*; Y)\, p(\pi^*)}{L(\pi^{(t)}; Y)\, p(\pi^{(t)})}$.
3. Move to π^* with probability $\min(1, \text{ratio})$. Otherwise stay at $\pi^{(t)}$ and record it again.

Because $p(Y)$ appears in both the numerator and the denominator, it cancels from the ratio and never has to be computed. Proposals that increase the posterior (ratio above 1) are always accepted, and proposals that decrease it are accepted only some of the time. The chain therefore spends more time where the posterior is high, and the recorded values are draws from the posterior. In practice, run four chains from different starting values, discard the burn-in (the early draws before the chain reaches the posterior) and check $\hat{R}$, which compares the chains. Values below 1.05 suggest convergence. Stan uses Hamiltonian Monte Carlo, which uses the gradient of the log posterior to make longer moves.

9 Exercises for Part II

Unless stated otherwise, these exercises use the Hainan data ($r = 12$, $n = 15$), the Beta$(6, 4)$ prior and the Beta$(18, 7)$ posterior.

Exercise 9. Does the likelihood integrate to one?

Setup. For Hainan, $L(\pi; Y) = \pi^{12}(1-\pi)^3$. A probability density for π must integrate to one over $[0, 1]$. Figure 3 includes a curve labelled "likelihood (rescaled to area one)".

Formulas. For integers a and b, $\int_0^1 \pi^{a-1}(1-\pi)^{b-1}\, d\pi = B(a, b) = \dfrac{(a-1)!\,(b-1)!}{(a+b-1)!}$.

Task. Is $\int_0^1 L(\pi; Y)\, d\pi = 1$? Compute the integral and explain what the rescaled curve in Figure 3 is.

No. With $a - 1 = 12$ and $b - 1 = 3$, $\int_0^1 \pi^{12}(1-\pi)^3 d\pi = B(13, 4) = \dfrac{12!\,3!}{16!} = \dfrac{6}{13 \cdot 14 \cdot 15 \cdot 16} = \dfrac{6}{43{,}680} \approx 1.37 \times 10^{-4}$. L is a function of π with the data held fixed, not a probability distribution for π (L6). Each value gives the probability of the observed data if π took that value. For a fixed π, the probabilities of all possible data sets sum to one, but the likelihood of one data set need not integrate to one across values of π.

Dividing L by its integral gives a density proportional to $\pi^{12}(1-\pi)^3$, which is the Beta$(13, 4)$ density. This is also the posterior under a flat Beta$(1, 1)$ prior, because in that case prior times likelihood is proportional to the likelihood. The rescaling allows Figure 3 to show the prior, likelihood and posterior on the same axes.

Exercise 10. A five-point grid (L13)

Setup. Suppose π can take only the five values 0.5, 0.6, 0.7, 0.8 and 0.9. The prior column gives the Beta$(6, 4)$ density at each value, rescaled to sum to one, and $L(\pi; Y) = \pi^{12}(1-\pi)^3$ for Hainan.

Formulas. On a grid, Bayes' rule has three steps. Multiply the prior by the likelihood in each row, sum the products to obtain $p(Y)$ for the grid, and divide each product by the sum.

Task. Complete the columns "Prior $\times L$" and "Posterior". Then compute the posterior under a flat prior (0.2 at each value). Where does each posterior reach its maximum, and why do the two differ?

π	$L(\pi; Y)$	Prior	Prior $\times L$	Posterior	Posterior, flat prior
0.5	3.1×10^{-5}	0.23	7.2×10^{-6}	0.03	0.02
0.6	1.4×10^{-4}	0.30	4.2×10^{-5}	0.17	0.10
0.7	3.7×10^{-4}	0.27	1.0×10^{-4}	0.41	0.27
0.8	5.5×10^{-4}	0.16	8.7×10^{-5}	0.35	0.40
0.9	2.8×10^{-4}	0.04	1.0×10^{-5}	0.04	0.21
Sum		1.00	2.5×10^{-4}	1.00	1.00

Multiply in each row. At $\pi = 0.7$, for example, $3.7 \times 10^{-4} \times 0.27 = 1.0 \times 10^{-4}$. The products sum to 2.5×10^{-4}, which is $p(Y)$ on this grid, and dividing by the sum gives posterior probabilities of 0.03, 0.17, 0.41, 0.35 and 0.04. Under a flat prior the 0.2 terms cancel, so the posterior is the likelihood divided by its sum (1.4×10^{-3}), which gives 0.02, 0.10, 0.27, 0.40 and 0.21.

With the Beta$(6, 4)$ prior the posterior is highest at 0.7, not at the MLE of 0.8. Fifteen respondents do not outweigh a prior worth 10 respondents and centred at 0.6. With the flat prior the posterior is the rescaled likelihood and is highest at the MLE (0.40 at 0.8, with 0.21 at 0.9). The Beta$(6, 4)$ prior moves probability from 0.9 (where it falls to 0.04) toward 0.6 and 0.7.

With a continuous flat prior the posterior is Beta$(13, 4)$. Its mode, $12/15 = 0.80$, equals the MLE, but its mean is 0.765 (the Beta$(1, 1)$ row of Table 5). A flat prior reproduces the MLE only for the posterior mode. The posterior also gives the plausibility of each value, whereas the MLE gives only the single best value.

Exercise 11. A conjugate update by hand

Setup. A small province has $n = 4$ respondents, and $r = 1$ of them trusts local government, so the MLE is $1/4 = 0.25$. Use the same Beta$(6, 4)$ prior as for Hainan (mean 0.60, worth 10 respondents).
Formulas. $\pi \mid Y \sim \text{Beta}(a + r,\ b + n - r)$, and the posterior mean written as a weighted average (§5).
Task. (a) What is the posterior distribution? (b) Write the posterior mean as a weighted average of the prior mean and the MLE. (c) Which receives more weight, and what follows from this?

(a) Beta$(6 + 1,\ 4 + 3) = \text{Beta}(7, 7)$, with mean $7/14 = 0.50$.
(b) The prior weight is $10/14 = 0.71$ and the data weight is $4/14 = 0.29$, so the posterior mean is $0.71 \times 0.60 + 0.29 \times 0.25 = 0.43 + 0.07 = 0.50$.
(c) The prior receives 71% of the weight, because four respondents carry less than half as much information as the prior. The posterior mean of 0.50 is 25 points above the MLE of 0.25, and the whole difference comes from the prior. For small provinces the result therefore depends heavily on the prior, which needs to be justified. A hierarchical model (L14) addresses this by estimating how much information to borrow from the other provinces (partial pooling).

Exercise 12. Confidence and credible intervals

Setup. Hainan has two 95% intervals for π: the Wald confidence interval $[0.598, 1.002]$ from ML and the credible interval $[0.53, 0.87]$ from the Beta$(18, 7)$ posterior. Under ML the data are random across repeated surveys. Under Bayesian inference π is fixed, and uncertainty about it is described by a probability distribution.
Task. For each statement, indicate whether it correctly describes each interval (Yes or No), and explain.

	Statement	Confidence interval	Credible interval
a	"Over repeated surveys, 95% of intervals built this way cover π."	Yes	No (not its definition)
b	"Given our model and prior, π lies in the interval with probability 0.95."	No	Yes
c	"The probability that more than 60% of Hainan adults trust is 0.90."	No: π is not random	Yes: $P(\pi > 0.6 \mid Y)$
d	"The interval can extend beyond $[0, 1]$."	Yes: Wald, $n = 15$	No

(a) This is the definition of a confidence interval, which is a property of the procedure over repeated samples. A credible interval is not constructed to have this property, and if the prior is poorly chosen its coverage across repeated samples can be below 95%.
(b) This is the definition of a credible interval. It does not apply to the confidence interval, because after the data are observed the fixed π is either inside the interval or not.
(c) $P(\pi > 0.6 \mid Y) = 0.90$ is computed from the posterior. ML does not give a probability distribution for π, so it cannot support this statement.
(d) The Wald interval, $\hat{\pi} \pm 1.96\,\text{se}$, is symmetric and does not respect the bounds of a probability. With $\hat{\pi} = 0.80$ and $n = 15$ it reaches 1.002. The posterior is a Beta distribution on $[0, 1]$, so the credible interval stays within that range.

Exercise 13. One Metropolis step

Setup. Hainan, Beta$(6, 4)$ prior, $r = 12$, $n = 15$. A Metropolis chain is at $\pi^{(t)} = 0.60$. The algorithm proposes a value π^*, computes the ratio of prior times likelihood at π^* to its value at $\pi^{(t)}$, and moves with probability $\min(1, \text{ratio})$ (§8).
Formulas. A Beta(a, b) density is proportional to $\pi^{a-1}(1 - \pi)^{b-1}$. Also, $\log(7/6) = 0.154$, $\log(3/4) = -0.288$ and $\log(2/3) = -0.405$.
Task. (a) Show that $L(\pi; Y)\, p(\pi) \propto \pi^{17}(1 - \pi)^6$. (b) What happens if the proposal is $\pi^* = 0.70$, and what happens if it is $\pi^* = 0.40$? Decide first without a calculator, then compute the ratios. (c) Why is $p(Y)$ not needed?

(a) $L \propto \pi^{12}(1 - \pi)^3$ and the Beta$(6, 4)$ density is proportional to $\pi^5(1 - \pi)^3$. Multiplying adds the exponents and gives $\pi^{17}(1 - \pi)^6$, the kernel of the Beta$(18, 7)$ distribution obtained from the conjugate formula. Its mode is $17/23 = 0.739$.
(b) The function increases up to the mode at 0.739 and decreases after it. The proposal 0.70 lies between 0.60 and the mode, so the ratio is above 1 and the chain always moves. The proposal 0.40 is farther from the mode on the same side, so the ratio is below 1 and the chain moves with probability equal to the ratio. For 0.70 the ratio is $(0.70/0.60)^{17}(0.30/0.40)^6 = \exp\{17(0.154) +$

$6(-0.288)\} = e^{0.894} = 2.45$. For 0.40 it is $(0.40/0.60)^{17}(0.60/0.40)^6 = (2/3)^{11} = \exp\{11(-0.405)\} = e^{-4.460} = 0.012$. The chain moves to 0.40 with probability 0.012 (draw $u \sim U(0,1)$ and move if $u < 0.012$) and otherwise records 0.60 again.

(c) $p(Y)$ is the same constant in the numerator and the denominator of the ratio, so it cancels. This is why MCMC can be used when $p(Y)$ cannot be computed.

Exercise 14. Priors on coefficients

Setup. This exercise returns to the separation problem in §7. One respondent in Hainan is aged 60 or older and she trusts local government, so the MLE of β_{60+} is $+\infty$. Two priors are considered, a weak $N(0, 2.5^2)$ prior and an informative $N(0.55, 0.5^2)$ prior based on the other provinces. The results are in the table in §7.
Task. (a) What does a $N(0, 2.5^2)$ prior on a logit coefficient rule out? (b) Why is the posterior for β_{60+} finite even though only one respondent is 60 or older, and why is its interval almost as wide as the prior's? What does the informative prior gain, and at what cost?

(a) About 95% of the prior probability lies within two standard deviations of zero, that is, within ± 5 on the log-odds scale. This corresponds to odds ratios between $e^{-5} \approx 1/148$ and $e^5 \approx 148$. The prior rules out only implausibly large effects.

(b) The posterior is proportional to the likelihood times the prior. As β_{60+} increases, the likelihood rises but levels off, while the normal prior density keeps decreasing toward zero, so the product has a maximum at a finite value (posterior mean 0.95). One respondent provides very little information, so the posterior is close to the prior. Its interval, $[-3.00, 5.43]$, is almost as wide as the central 95% range of the prior, about ± 4.9. The informative prior narrows the interval to $[-0.32, 1.65]$ around the true value of 0.55. The cost is that this information comes from outside the Hainan data, here from the other provinces, and the choice has to be justified.

Exercise 15. Errors in interpretation (Part II)

Setup. As in Exercise 8, each sentence contains one error, here about the likelihood, the prior or an interval.
Task. Explain the error and write a corrected sentence.

	Sentence	Error and correction
1	"The posterior is the likelihood."	The posterior is proportional to the likelihood times the prior. The likelihood is not a distribution over π (its integral is 1.37×10^{-4}; Exercise 9). The posterior equals the rescaled likelihood only under a flat prior.
2	"ML is objective; Bayes is subjective."	Both approaches rely on assumptions. ML assumes the model, a random sample and, for Wald intervals, a large n. The prior is an additional assumption that is stated explicitly and can be checked with a sensitivity analysis (L13).
3	"With a flat prior, Bayes is ML."	Only the posterior mode equals the MLE (0.80). The posterior mean is 0.765, and the two intervals have different interpretations.
4	"Metropolis needs $p(Y)$."	Metropolis requires only prior times likelihood at two values. $p(Y)$ appears in both the numerator and the denominator of the ratio and cancels.
5	"A prior worth 10 means we invented 10 respondents."	The worth of a prior measures how much information it carries, in units of respondents. Its weight, $(a+b)/(a+b+n)$, declines as n increases: 40% for Hainan and 4% for Guangdong.
6	"A 95% confidence interval and a 95% credible interval are the same thing."	The two intervals make different probability statements. For the confidence interval, 95% of intervals from repeated samples contain π. For the credible interval, the probability that π lies in the interval is 0.95, given the model and prior. For Hainan they are $[0.598, 1.002]$ and $[0.53, 0.87]$.

Appendix A Pocket card

PART I Reading coefficients

GLM (L7). $Y_i \sim$ dist., $\eta_i = x_i^\top \beta$, $g(\mathrm{E}[Y_i \mid x_i]) = \eta_i$. One unit of x_{im} adds β_m to η_i. **Routine:** sign $\to e^{\hat\beta_m} \to e^{c\hat\beta_m}$ (doubling: $c = \log 2$) $\to$ predicted quantity if you need probabilities, counts or survival. Interpret $\hat\beta$, not β; ceteris paribus.

OLS (L2). $\hat\beta = (X^\top X)^{-1} X^\top y$; FWL: slope of $\tilde y$ on $\tilde x$; $\beta^\dagger = \mathrm{E}[XX^\top]^{-1}\mathrm{E}[XY]$ is the best linear predictor. Identity link: $\hat\beta_m$ is the effect.

LPM. $P(Y = 1 \mid x) = x^\top \beta$; $\hat\beta$ in probability points; can leave $[0, 1]$; heteroskedastic: use robust se.

Logit. $\partial \log\{\pi/(1-\pi)\}/\partial x_j = \beta_j$; $e^{\hat\beta_j}$ = odds ratio; $\hat\pi(x) = \mathrm{logit}^{-1}(x^\top\hat\beta) = e^z/(1+e^z)$; $\partial\hat\pi/\partial x_j = \hat\beta_j \hat\pi(1-\hat\pi)$. Contrast $\hat\tau_j(x) = \hat\pi(x_j{=}a, x_{-j}) - \hat\pi(x_j{=}b, x_{-j})$. PEA: at the average x. APE: $\frac{1}{n}\sum_i \hat\tau_i$ (prefer). Probit: $F = \Phi$; compare $\hat\pi$, not $\hat\beta$.

Uncertainty for $T(\hat\beta)$. $\hat\beta \overset{a}{\sim} N(\beta, [-H]^{-1})$; draw $\beta^{(k)} \sim N(\hat\beta, \widehat{\mathrm{Var}}(\hat\beta))$, compute $T(\beta^{(k)})$; se = sd, interval = 2.5/97.5 percentiles; or delta method.

Ordered logit (L9). $Y^* = x^\top\beta + \varepsilon$, cutpoints ζ_j (no intercept). $P(Y \le j \mid x) = \mathrm{logit}^{-1}(\zeta_j - x^\top\beta)$; $P(Y = j) = \mathrm{logit}^{-1}(\zeta_j - x^\top\beta) - \mathrm{logit}^{-1}(\zeta_{j-1} - x^\top\beta)$. $\beta_m = \partial \log \frac{P(Y>j)}{P(Y\le j)}/\partial x_m$, same for all j; $e^{\hat\beta_m}$ multiplies odds of being above any cutpoint. Middle categories: direction ambiguous. APEs sum to 0.

Multinomial (L9). $P(Y = j \mid x) = \exp(x^\top\beta_j)/\sum_k \exp(x^\top\beta_k)$, $\beta_1 = 0$. $\log\frac{P(j)}{P(1)} = x^\top\beta_j$; $e^{\hat\beta_{jm}}$ multiplies $P(j)/P(1)$. $\frac{P(j)}{P(k)} = \exp\{x^\top(\beta_j - \beta_k)\}$ (IIA). Coefficient sign $\neq$ sign of APE.

Poisson, NB (L10). $\mathrm{E}[Y \mid x] = \mu = \exp(x^\top\beta)$; $e^{\hat\beta_m}$ = IRR (ratio, not difference); gap $= \hat\mu(e^{\hat\beta_m} - 1)$. Poisson: Var $= \mu$; NB: Var $= \mu + \mu^2/\theta$. Overdispersion $\Rightarrow$ se too small: quasi-Poisson ($\phi\mu$), sandwich.

Hazard (L12). $\log\{-\log(1-h_{ij})\} = \gamma_j + x_i^\top\beta$; $e^{\hat\beta_m}$ = hazard ratio (same in every j). KM: $\hat h_j = q_j/r_j$, $\hat S(t) = \prod_{j\le t}(1-\hat h_j)$; $\hat S(t \mid x) = \prod(1-\hat h_j(x))$; $S(t \mid x_1) = S(t \mid x_0)^{e^\beta}$ when one unit apart. Censoring: independent.

Hierarchical (L14). $\beta_{0j} = \mu + u_j$, $u_j \sim N(0, \sigma_u^2)$; $\hat\beta$ hold u_j fixed. $\hat\pi_c = \mathrm{logit}^{-1}(x_c^\top\hat\beta + \hat u_j)$; MRP: $\hat\theta_j = \sum_{c\in j} s_c \hat\pi_c$. Fixed effects: nuisance; random: levels.

Penalized (L5). Ridge $\hat\beta^R_\lambda = (X^\top X + \lambda I)^{-1}X^\top y$; lasso $\lambda\|b\|_1$ gives zeros; both biased; λ by CV.

Values. $\mathrm{logit}^{-1}(z)$ at $z = 0, 1, 2, 3$: 0.50, 0.73, 0.88, 0.95; for $-z$ take $1-$ these. $e^{0.5}$=1.65, e=2.72, e^2=7.39, e^{-1}=0.37, $\log 2$=0.69.

PART II Likelihood and Bayes

Likelihood (L6). $L(\theta; Y) = p(Y \mid \theta)$: a function of θ, data fixed; *not* a distribution for θ. $\ell = \log L$, $\hat\theta = \arg\max \ell$. Score $s = \partial\ell/\partial\theta$; Hessian H; $-H$ large $\Rightarrow$ spiky, informative. Newton–Raphson: $\theta^{(t+1)} = \theta^{(t)} - H(\theta^{(t)})^{-1}s(\theta^{(t)})$. $-\ell/n$ is the average log loss; normal $Y \Rightarrow$ OLS is ML.

Bernoulli. $L = \pi^r(1-\pi)^{n-r}$; $s = \frac{r}{\pi} - \frac{n-r}{1-\pi}$; $\hat\pi = r/n$; se $= \sqrt{\hat\pi(1-\hat\pi)/n}$; Wald $\hat\pi \pm 1.96$ se.

Logit ML. $s(\beta) = \sum_i (y_i - \pi_i)x_i$; $-H = X^\top W X$, $W = \mathrm{diag}\{\pi_i(1-\pi_i)\}$; $\widehat{\mathrm{Var}}(\hat\beta) = (X^\top W X)^{-1}$; assumes the model is true and n large. Separation: MLE $= \pm\infty$.

ML uncertainty. π fixed, *data* random across repeated samples. 95% CI: the 95% describes the *procedure*.

Bayes (L13). $p(\pi \mid Y) = \dfrac{L(\pi; Y)\, p(\pi)}{p(Y)} \propto L(\pi; Y)\, p(\pi)$; $p(Y) = \int L\, p\, d\pi$ only normalizes. π still fixed; the prior and posterior describe *our uncertainty*.

Beta–Binomial. $\pi \sim \mathrm{Beta}(a, b)$, $\mathrm{E}\pi = \frac{a}{a+b}$, worth $a + b$. $\pi \mid Y \sim \mathrm{Beta}(a + r, b + n - r)$. $\mathrm{E}[\pi \mid Y] = \frac{a+b}{a+b+n}\frac{a}{a+b} + \frac{n}{a+b+n}\frac{r}{n}$. A prior matters when n is small.

Summaries. Posterior mean/median/mode; 95% credible interval: $P(\pi \in \mathrm{CrI} \mid Y) = 0.95$ given model and prior; $P(\pi > c \mid Y)$.

Priors on coefficients. $\beta_j \sim N(m_j, s_j^2)$; weak $N(0, 2.5^2)$ rules out OR $> e^5 \approx 148$. Choose: let data swamp it; check sensitivity; weak; estimate it (hierarchical, empirical Bayes: partial pooling, shrinkage toward $\hat\mu$).

Approximating a posterior. Grid (100^k evaluations). Metropolis: $\pi^* = \pi^{(t)} + \varepsilon$, $\varepsilon \sim N(0, s^2)$; ratio $= \frac{L(\pi^*)p(\pi^*)}{L(\pi^{(t)})p(\pi^{(t)})}$; accept with prob. $\min(1, \text{ratio})$, else stay; $p(Y)$ cancels. Burn in; 4 chains; $\hat R < 1.05$; Stan uses HMC. Markov chain + Monte Carlo = MCMC.

Hainan. $r = 12$, $n = 15$: $\hat\pi = 0.80$, se 0.103, CI $[0.598, 1.002]$. $\mathrm{Beta}(6, 4) \to \mathrm{Beta}(18, 7)$: mean $0.72 = 0.40(0.60) + 0.60(0.80)$, CrI $[0.53, 0.87]$, $P(\pi > 0.6 \mid Y) = 0.90$. Flat prior: $\mathrm{Beta}(13, 4)$, mode 0.80, mean 0.765.

Model comparison (L10). $LR = 2\{\ell(\text{with}) - \ell(\text{without})\} \sim \chi^2_k$ (nested only); AIC $= -2\hat\ell + 2d$; BIC $= -2\hat\ell + d\log n$.

Five parts of an interpretation: quantity and contrast; size of change; values of other x (PEA/APE); scale; uncertainty.

Sources and reproducibility. Every coefficient and table entry quoted from a lecture is the value printed on the slides (GOVT 6039, Lectures 2 and 5–14; github.com/ddekadt/cornell-govt-6039). Every derived number (predicted probabilities, odds, gaps, grid and posterior values, figures) is recomputed in `compute.py` from those printed estimates, and checked against every number the slides themselves print (61 checks). Because coefficients are printed to two decimals, a recomputed number can differ from a slide figure in the last digit. The hypothetical turnout logit (Exercise 3), the baseline survival of 0.50 (Exercise 7) and the small-province update (Exercise 11) are illustrations, not lecture data.